

Henrique Soares Assumpção e Silva

Machine Learning Researcher

Belo Horizonte, Brazil

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🌐 [Website](#) 🐙 [Github](#)

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Education

Universidade Federal de Minas Gerais (UFMG)

M.S. in Computer Science; GPA: 99%

Mar 2025 – Jul 2026

Belo Horizonte, Brazil

- **Thesis:** [Semidefinite Optimization on Laplacians of Highly Regular Graphs](#)
- **Advisor:** [Prof. Gabriel Coutinho](#)

Universidade Federal de Minas Gerais (UFMG)

B.S. in Computer Science, Minor in Mathematics; GPA: 94%

Mar 2020 – Dec 2024

Belo Horizonte, Brazil

- **Thesis:** [Algebras, groups and graphs](#)
- **Advisor:** [Prof. Gabriel Coutinho](#)

Professional Experience

Machine Learning Research Scientist

[Inter Bank](#)

Jun 2025 - Ongoing

Belo Horizonte, Brazil

- Lead an interdisciplinary team on the *Large Transaction Models* project, directing the pre-training and optimization of 10B+ parameter foundation models on large-scale financial datasets across multi-node GPU clusters for downstream credit risk analysis and financial forecasting.
- Created, open-sourced, and led the development of [CodeEvolve](#), an evolutionary coding agent for scientific discovery. CodeEvolve outperforms other state-of-the-art frameworks, including DeepMind's AlphaEvolve, on multiple code generation benchmarks.

Machine Learning & Optimization Research Fellow

[Department of Computer Science - UFMG](#)

Dec 2020 - May 2025

Belo Horizonte, Brazil

- Developed and deployed a Variational Autoencoder in PyTorch for predictive maintenance on industrial machinery, improving accuracy over legacy systems by 10%.
- Designed a recurrent neural network to predict emotional shifts across large online communities, outperforming established baselines by 20%.
- Derived novel semidefinite programming bounds for NP-hard combinatorial problems using spectral graph theory.

Data Science Instructor

[Usiminas](#)

Jun 2022 - Dec 2022

Belo Horizonte, Brazil

- Designed and delivered hands-on instruction in Python, Pandas, and PyTorch to engineering teams at a major steel manufacturer.
- Mentored internal teams building ML tools, contributing to the automation of legacy data pipelines.

Machine Learning Research Intern

[Inter Bank](#)

Aug 2021 - Jun 2022

Belo Horizonte, Brazil

- Developed *DELATOR*, a Graph Neural Network framework for modeling large financial transaction networks with 20M+ nodes and 100M+ edges.
- Deployed the pipeline into production banking environments, improving money laundering detection over legacy baselines.

A.I. Research & Development Engineer

[Plus Three](#)

Mar 2021 - Aug 2021

Remote

- Built NLP pipelines using Transformer-based models for multi-turn question answering and text generation in production web systems.
- Wrote a technical advisory series on algorithmic bias and fairness for the non-profit *AlandYou*.

Academic Service

Committee Member

[AIDDA Institute](#)

Mar 2026 - Ongoing

Remote

- Helps coordinate an international network of researchers working on recursive self-improvement and agent-driven scientific discovery to align methodologies and open challenges.
- Facilitates global community initiatives and cross-institutional discussions, including contributing to the organization and technical program of the inaugural AIDDA conference.

Papers

Conference Proceedings

- **Henrique Assumpção**, Diego Ferreira, Leandro Campos, Fabricio Murai. [CodeEvolve: An Open-Source Evolutionary Coding Agent for Algorithmic Optimization](#). *Findings of the Association for Computational Linguistics (EMNLP 2026)*.
- **Henrique Assumpção**, Fabrício Souza, Leandro Lacerda Campos, Vinícius T. de Castro Pires, Paulo M. Laurentys de Almeida, Fabricio Murai. [DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs](#). *IEEE International Conference on Big Data (IEEE BigData 2022)*.

Journal Articles

- **Henrique Assumpção**, Gabriel Coutinho, Chris Godsil. [Total Conformal Rigidity in Graphs](#). Under revision at *Journal of Graph Theory*, 2026.
- **Henrique Assumpção**, Gabriel Coutinho. [Semidefinite programming bounds on fractional cut-cover and maximum 2-SAT for highly regular graphs](#). *Mathematics of Operations Research*, 2026.
- Bárbara Silveira*, **Henrique Assumpção***, Fabricio Murai, Ana Paula C. da Silva. [Predicting user emotional tone in mental disorder online communities](#). *Future Generation Computer Systems*, 2021.

*Equal contribution.

Talks and Presentations

- [CodeEvolve: an open source evolutionary coding agent for algorithmic discovery and optimization](#). Poster Presentation. *EMNLP 2026*. Budapest, Hungary.
- [CodeEvolve: Evolutionary Agents for Scientific Discovery](#). Invited Talk. *AIDDA 2026*. Remote.
- [CodeEvolve: an Open Source Evolutionary Coding Agent for Algorithmic Discovery and Optimization](#). Invited Talk. *Inter Science Talks 2026*. Belo Horizonte, Brazil.
- [Gauge duality for parameters of highly regular graphs](#). Invited Talk. *ILAS 2026*. Blacksburg, USA.
- [Evolutionary Coding Agents](#). Invited Talk. *Automated Discovery at Scale 2026*. San Francisco, USA.
- [MAXCUT, Association Schemes and Semidefinite Programs](#). Invited Talk. *CNMAC 2025*. Rio de Janeiro, Brazil.
- [Primal-dual inequalities for the approximate maximum-cut value of Graphs in Homogeneous Coherent Configurations](#). Poster Presentation. *CNMAC 2024*. Porto de Galinhas, Brazil.
- [DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs](#). Oral Presentation. *IEEE BigData 2022*. Osaka, Japan.

Technical Skills

Programming Languages: C, C++, Python

Tools & Technologies: Apache Spark, AWS Sagemaker, CUDA, Docker, Git, HuggingFace, Megatron, Pytorch

Languages: Portuguese (Native), English (C2), Spanish (B2), French (B1)